(1) $\det(2A) = 2^n \times \det(A) = 2^4 \times \frac{1}{2} = 8$

$\det(A^{-1}) = \frac{1}{\frac{1}{2}} = 2$

$\det(A^2) = (\det(A))^2 = (\frac{1}{2})^2 = \frac{1}{4}$

if Multiply a mat with -1,

det becomes $(-1)^n \times \det(A)$

$\therefore \det(-A) = (-1)^4 \times \frac{1}{2} = \frac{1}{2}$

(2) $\det \frac{1}{2}A = (\frac{1}{2})^3 \times (-1) = -\frac{1}{8}$

$\det(-A) = (-1)^3 (-1) = 1$

$\det(A^2) = (-1)^2 = 1$

$\det(A^{-1}) = -\frac{1}{1} = -1$

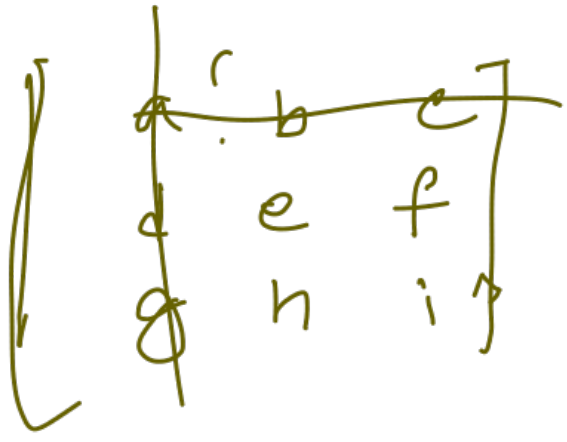
$\therefore \det(A) = 0$, then all would be

zero and no A

(3) a) 3 → 1
4 → 2
5 → 2
6 →

(4)

(5)



$a(ei-fh)$

(13) skew sym of odd order → det = 0

For " " $A^T = -A$

$$\det(A^T) = \det(-A) \quad \text{—— (1)}$$

$$\det(-A) = (-1)^n \det(A)$$

if n is odd, $\det(-A) = -\det(A)$ —— (2)

(1) (2)

$\det(A^T) = \det(A)$ → same coeff as col in row of A^T

so, (0) =>

$\det(A) = -\det(A)$

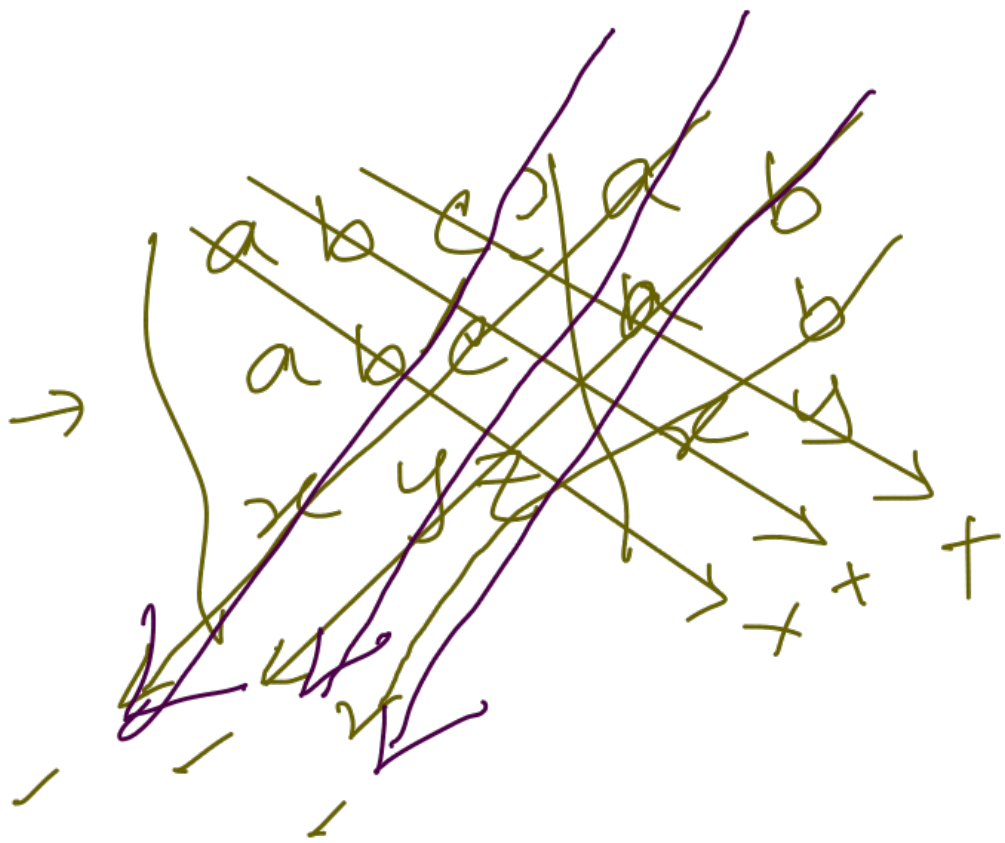
$\therefore 2\det(A) = 0 \therefore \det(A) = 0$

(15)

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{\min} \quad \begin{bmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & d \end{bmatrix}$$

a b c
p q r
x y z

→



$abz + bcx + cay - ab^2$

$acy - cbx$

$= 0$